

Ajeenckya Mahadik

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Skills

Languages & Backend: Python, SQL, SQLite, FastAPI, Streamlit, Playwright, Git, REST APIs.

ML & Deep Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, Bayesian Modeling, LoRA/QLoRA Fine Tuning.

LLM & Agents: LLMs, RAG, Knowledge Graph RAG, Tool Use, Agent Memory, ReAct, Guardrails, LangChain, LangGraph, Multi-agent Orchestration, RLAIIF, Hugging Face, LLM Evaluation.

Data & Cloud: PostgreSQL, Redis, pgvector, AWS S3, Lambda, Docker, Kubernetes, CI/CD, Model Deployment.

Work Experience

University of Wisconsin–Madison

Madison, WI

Graduate Research Assistant

May 2025 – Present

- Built a probabilistic forecasting stack with a PyTorch **policy gradient neural network** over multiple engineered features and an XGBoost weather sub model served via FastAPI, improving forecast accuracy by **12%**.
- Extended the policy gradient neural network with a **Bayesian neural network output head** via Monte Carlo sampling, reducing Brier score from 0.24 to **0.176**, a **26.7%** improvement over baseline, recall to **84%**, and variance by **17%**.
- Validated **neural network and XGBoost** variants using Brier score and **McNemar tests** via an A/B framework and logged calibration diagnostics and experiment metadata in PostgreSQL.

Fiat India Automobiles Pvt. Ltd.

India

Graduate Associate Engineer

Sept 2023 – June 2024

- Trained a **Gradient Boosting Regressor** on 150K assembly records to predict operation level delay, giving managers a 2 to 4 hour intervention window and lifting line efficiency by **9%**.
- Engineered queue depth, shift context, operator history, and product complexity features from assembly logs via schema standardization, temporal deduplication, and leakage aware splits.
- Provisioned an AWS pipeline over S3, Lambda, and CloudWatch with IAM access controls to deliver delay features and control plan outputs to manufacturing teams.

Powertrac

India

Analytics Engineer

July 2022 – July 2023

- Built a predictive maintenance **XGBoost classifier** on diagnostic logs with mean time between failure, fault recurrence, and failure frequency features to predict 30 day breakdown risk, cutting downtime by **12%**.
- Implemented an **XGBoost risk score** driven dispatch system using **greedy knapsack** technician assignment to replace manual triage, improving routing efficiency by **22%**.
- Established a **model monitoring** loop tracing low confidence predictions to upstream feature drift, sustaining precision above **88%** in production.

Tata Motors

India

Graduate Engineer

Feb 2021 – Feb 2022

- Consolidated cross system machine and tool logs by normalizing timestamps, de-duplicating throughput windows, and deriving lag and rolling statistics features for anomaly modeling.
- Deployed **Isolation Forest** with z score thresholds on throughput, error rate, and readiness signals to surface anomalies before cascading, improving efficiency by **18%** and throughput by **15%**.
- Modeled deployment impact via **Monte Carlo simulation** over historical process patterns, reducing operational costs by **20%** and winning the **Kaizen Award**.

Projects

Autonomous CLI Coding Agent

- Engineered a framework free LLM coding agent with raw API tool calling loops, **ReAct planning**, **session memory**, and **RLAIIF scoring** via Grok 4, achieving **100% task completion** and **7.5x lower latency** than LangChain on identical 15 task tool calling benchmark, containerized with **Docker**.

Self-Improving LLM Agent

- Built an agent with **Plan-Act-Observe** and **ReACT** as a baseline with **ChromaDB** strategy memory and Grok 4 distillation into a **QLoRA LLaMA-3.2-1B** model, achieving **94% task completion** on Tau Bench.

DocGraph

- Designed a **Knowledge Graph RAG** system on 10K emails using **LightRAG**, **pgvector**, **Mistral 7B**, and **JWT isolation**, where graph traversal enables multi-hop entity linking unavailable in flat vector search, and E5 **embedding caching** cut retrieval latency by **44%** from 1.8s to 1.0s.

Education

University of Wisconsin–Madison

Madison, WI

M.S. Industrial Engineering, *Specialization in AI/ML*

May 2026

Relevant Coursework: AI Agents, Advanced Deep Learning, Foundation Models, DL for NLP, Machine Learning

Shivaji University

India

B.E. Mechanical Engineering

May 2020